SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme : B. Tech IT

Course Code : ITA0201

Course Name: WEB TECHNOLOGY

ASSIGNMENT 1 – SCENARIO-BASED ASSIGNMENT

Max Marks: 100

Code of Conduct

I **KAVIYA M 192421050** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kaviya M

______________________ **Faculty In-charge** **Name:** **Date:**	______________________ **Course Coordinator** **Name:** **Date:**	______________________ **Head of Department** **Name:** **Date:**

Assignment Evaluation Rubric

Criterion	Marks
HTML/XHTML structure and semantic organization	**15**
Lists, tables, hyperlinks and registration form	**15**
CSS design, layout and responsiveness	**20**
JavaScript validation and event handling	**15**
Functions, arrays/objects and credit calculation	**15**
Dynamic registration summary	**10**
Debugging, code quality and execution	**10**
Total	**100**

CO Assessed

CO I: Create web pages using HTML/XHTML and XML, and apply HTTP and Internet protocols for web content delivery.

Topics covered: Client-side webpage structure, HTML/XHTML rules, lists, tables, hyperlinks, forms, GET/POST awareness, and creation of HTML documents.

CO II: Apply CSS techniques including layout design (Flexbox, Grid), typography, colour schemes, and media queries to create responsive and user-friendly web interfaces.

Topics covered: CSS selectors and styling, box model, layout, responsive design, JavaScript literals, functions, arrays, objects, built-in features, validation, and debugging.

ABSTRACT

The **University Course Registration Portal** is a responsive web application developed to provide students with an easy and interactive platform for viewing and registering for university courses. The system presents course information in an organized manner and allows students to enter their registration details and select suitable courses.

The application is developed using **HTML, CSS, and JavaScript**. HTML is used to create the webpage structure, including navigation, course tables, lists, forms, and contact information. CSS is used to create a professional interface using Flexbox, Grid, responsive design, spacing, typography, and hover effects. JavaScript provides client-side validation, course selection, credit calculation, and dynamic registration summary generation.

The system validates student information before processing the registration and automatically calculates the number of selected courses and total credits. A registration summary containing the student's details and selected courses is displayed without reloading the webpage.

The project also demonstrates the use of browser Developer Tools for debugging a JavaScript credit calculation issue. Overall, the portal demonstrates the practical application of fundamental Web Technology concepts in developing a responsive and interactive academic webpage.

1. INTRODUCTION

Course registration is an important activity for university students. Students need to view available courses, select subjects, and provide their academic information during registration.

A well-designed registration portal should provide clear information, easy navigation, responsive design, proper validation, and immediate feedback.

The **University Course Registration Portal** was developed to provide these features using HTML, CSS, and JavaScript.

Objectives

- Create a professional course registration webpage.
- Display available courses clearly.
- Provide a student registration form.
- Validate user input using JavaScript.

- Allow students to select multiple courses.
- Calculate total courses and credits automatically.
- Display a registration summary dynamically.
- Make the interface responsive.

2. PROBLEM STATEMENT

Basic registration webpages may contain poorly organized information and limited interaction. Students may also face problems when entering incorrect information or calculating their total course credits.

The proposed system solves these problems by providing:

- Organized course information.
- A user-friendly registration form.
- Client-side validation.
- Dynamic course selection.
- Automatic credit calculation.
- Responsive design.
- Instant registration summary.

3. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Standard processor
- Standard display

Software Requirements

- Windows/Linux/macOS
- Visual Studio Code or similar editor

- Google Chrome / Microsoft Edge
- HTML5-supported browser

Technologies Used

Technology	Purpose
HTML5	Webpage structure
CSS3	Styling and responsive design
JavaScript	Interaction and validation
Flexbox	Layout
CSS Grid	Structured layout
DOM	Dynamic content
Developer Tools	Debugging

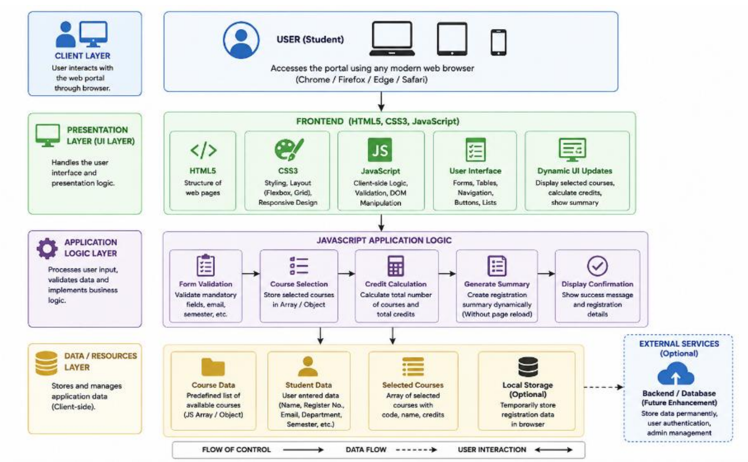
4. SYSTEM DESIGN

4.1 Architecture

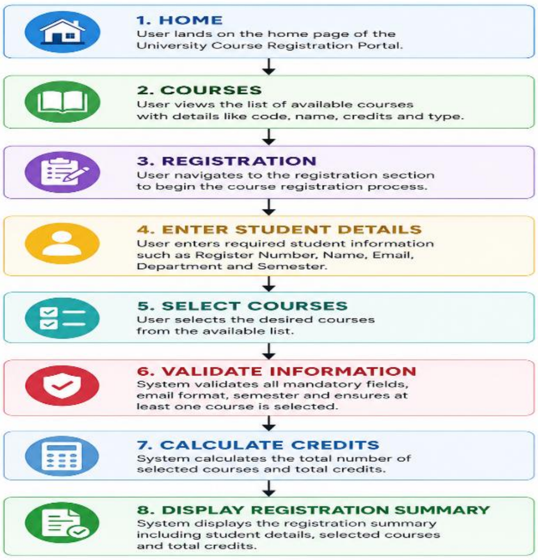
The system follows a simple client-side architecture.

The student interacts with the portal through a web browser. HTML creates the structure, CSS provides the design, and JavaScript performs validation and dynamic operations.

Architecture Flow



4.2 Registration Flow



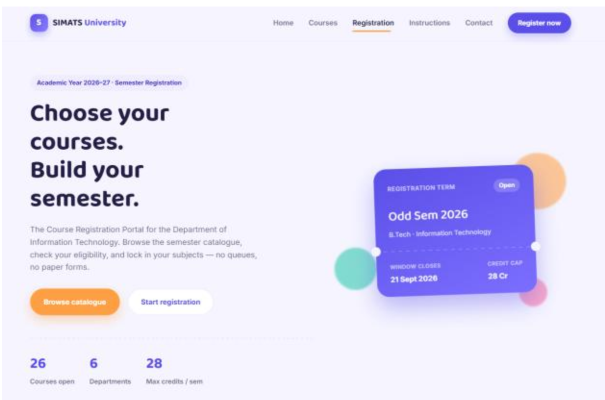
5. USER INTERFACE

5.1 Home Page

The home page introduces the University Course Registration Portal and provides navigation links to the main sections.

It contains:

- University title
- Navigation menu
- Introduction
- Registration section
- Contact information

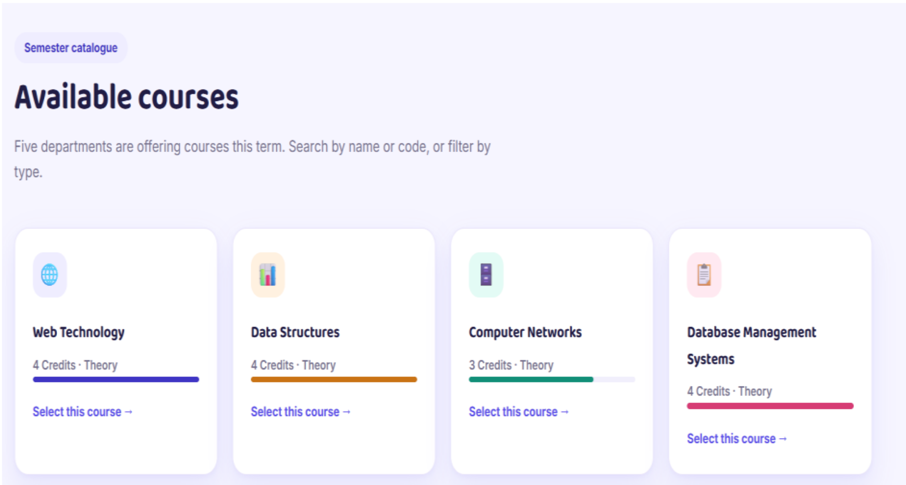


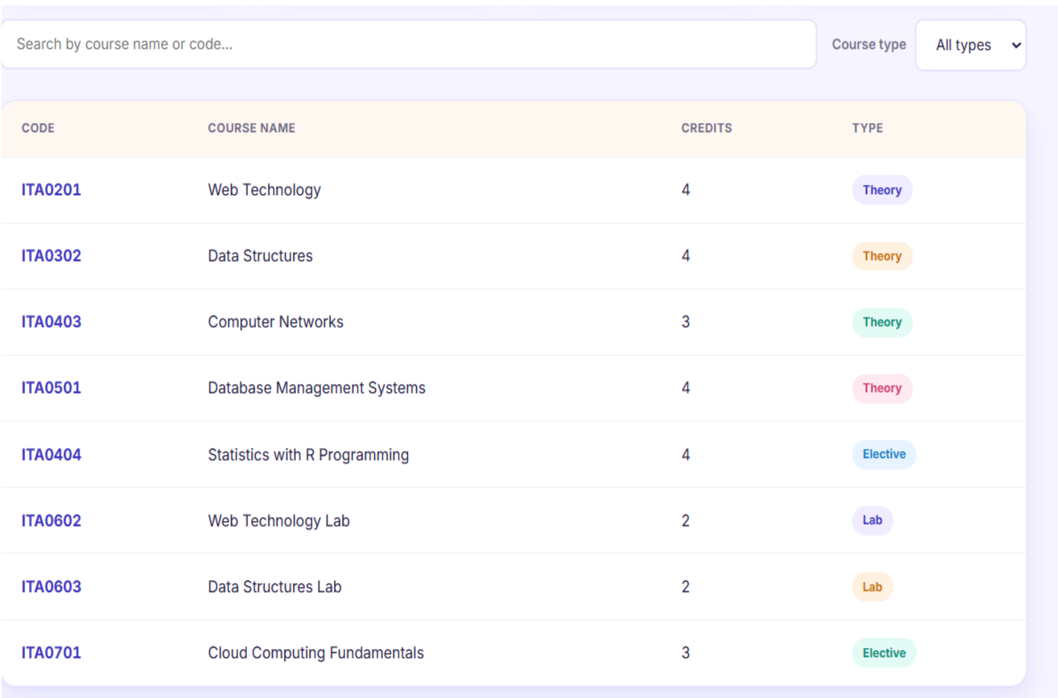
5.2 Course Section

The course section displays available subjects using a structured table.

The table contains:

- Course Code
- Course Name
- Credits
- Course Type





Search by course name or code...

Course type: All types

CODE	COURSE NAME	CREDITS	TYPE
ITA0201	Web Technology	4	Theory
ITA0302	Data Structures	4	Theory
ITA0403	Computer Networks	3	Theory
ITA0501	Database Management Systems	4	Theory
ITA0404	Statistics with R Programming	4	Elective
ITA0602	Web Technology Lab	2	Lab
ITA0603	Data Structures Lab	2	Lab
ITA0701	Cloud Computing Fundamentals	3	Elective

5.3 Navigation

The navigation menu provides links to:

Home | Courses | Registration | Contact

Internal hyperlinks allow users to move between different sections of the webpage.

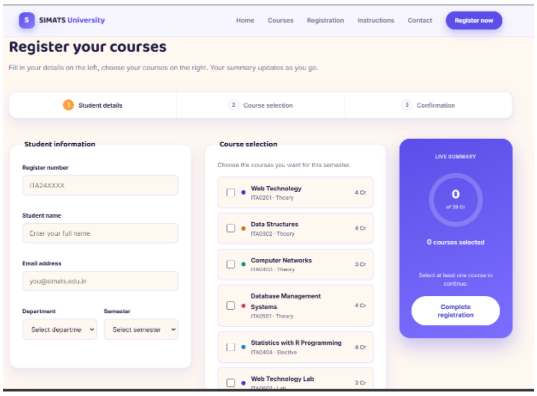
6. REGISTRATION MODULE

The registration module allows students to enter their basic academic information.

The form contains:

- Register Number
- Student Name
- Email
- Department
- Semester
- Course Selection

Students can select one or more courses from the available options.



7. JAVASCRIPT FUNCTIONALITY

JavaScript is responsible for making the portal interactive.

The major functions include:

- Form validation
- Course selection

- Course removal
- Course counting
- Credit calculation
- Registration summary generation
- Event handling
- DOM manipulation

7.1 Form Validation

The system checks whether the required fields are filled correctly.

It also verifies:

- Valid email format
- Appropriate semester
- At least one selected course

Invalid information produces an appropriate error message.

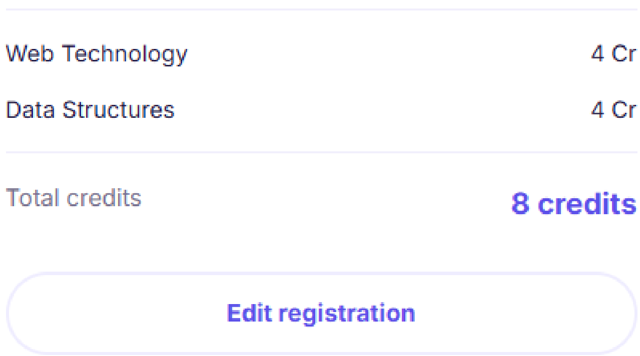
7.2 Course Selection

Course information is stored using JavaScript arrays and objects. When a student selects a course, the selected course information is updated dynamically.

7.3 Credit Calculation

The system automatically calculates the total credits.

For example:

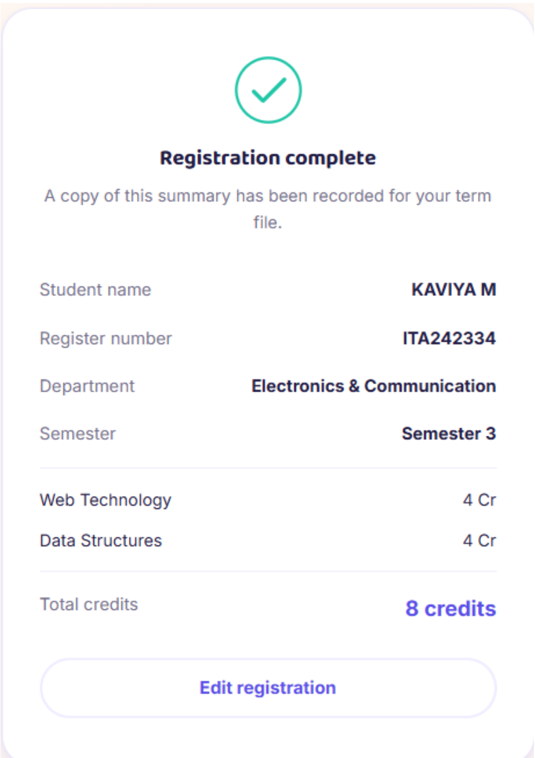


7.4 Registration Summary

After successful validation, the system displays:

- Student Name
- Register Number
- Department
- Selected Courses
- Total Courses
- Total Credits

The summary appears without reloading the webpage.



8. CSS AND RESPONSIVE DESIGN

CSS is used to create the professional appearance of the portal.

The project applies:

- CSS selectors
- Box model

- Margins and padding
- Borders and border radius
- Typography
- Flexbox
- CSS Grid
- Hover effects
- Media queries

Flexbox and Grid are used to organize page elements, while media queries make the webpage responsive.

9. TESTING

The portal was tested using different inputs and user actions.

Test	**Expected Result**	**Status**
Open webpage	Home page displayed	Pass
Navigate to Courses	Course section displayed	Pass
Empty form submission	Error message	Pass
Invalid email	Validation message	Pass
Select course	Course added	Pass
Select multiple courses	Course count updated	Pass
Select courses	Credits calculated	Pass
Valid registration	Summary displayed	Pass
Mobile screen	Responsive layout	Pass

The testing confirms that the major functions of the application work as expected.

10. DEBUGGING

Browser Developer Tools were used to identify and correct errors during development.

One JavaScript issue occurred because the course object used **credit**, while the calculation function expected **credits**.

Using console.log() helped inspect the course object and identify the mismatch.

After changing the property to **credits**, the total credit calculation worked correctly.

```
Registration attempt:                                          script.js:356
 ▾{regNumber: 'ITA242343', studentName: 'KAVIYA M', email: 'kaviya@simats.e
  du.in', department: 'IT', semester: '4', …} i
    department: "IT"
    email: "kaviya@simats.edu.in"
    regNumber: "ITA242343"
  ▸ selectedCourses: (2) ['ITA0302', 'ITA0403']
    semester: "4"
    studentName: "KAVIYA M"
  ▸ [[Prototype]]: Object
valid: true
> ctrl i to turn on code suggestions. Don't show again NEW
```

11. RESULTS

The completed portal successfully provides an interactive course registration experience.

The system allows students to:

- View available courses.
- Enter registration details.
- Select courses.
- Receive validation messages.
- View selected courses.
- Calculate total credits.
- Generate a registration summary.

The responsive design also allows the portal to work effectively on different screen sizes.

12. ADVANTAGES

The main advantages are:

1. Simple and attractive interface.
2. Easy navigation.
3. Responsive design.
4. Immediate form validation.
5. Automatic credit calculation.
6. Dynamic registration summary.
7. Interactive course selection.
8. Easy-to-maintain frontend structure.

13. LIMITATIONS

The current project is mainly a frontend application.

Its limitations include:

- No database connectivity.
- No student login system.
- No permanent registration storage.
- Static course information.
- No administrator dashboard.
- No server-side validation.

14. FUTURE ENHANCEMENTS

The portal can be extended by adding:

- Student authentication.
- MySQL database integration.
- Admin dashboard.

- Course management.
- Course seat availability.
- Registration history.
- Automatic timetable generation.
- Email confirmation.

These features can transform the project into a complete university course management system.

15. CONCLUSION

The **University Course Registration Portal** successfully demonstrates the practical implementation of fundamental Web Technology concepts in a real-world academic scenario. The project provides students with a simple, organized, responsive, and interactive platform for viewing available courses and completing their registration details.

The portal combines **HTML, CSS, and JavaScript** effectively. HTML is used to create the semantic structure of the webpage, including navigation links, course tables, lists, forms, and contact information. CSS improves the overall appearance through proper spacing, typography, box model properties, Flexbox, Grid, hover effects, and responsive design. JavaScript adds dynamic functionality such as form validation, course selection, credit calculation, event handling, and DOM manipulation.

One of the important features of the project is the dynamic registration summary. After entering valid information and selecting courses, students can immediately view their registration details, selected courses, total number of courses, and total credits without reloading the webpage.

The project also provided practical experience in testing and debugging. A JavaScript property mismatch involving credit and credits was identified using the browser Developer Console and corrected successfully. This helped demonstrate the importance of debugging while developing interactive web applications.

Overall, the project meets the major requirements of the Web Technology assignment and provides a strong foundation for further development. In the future, the portal can be extended with database connectivity, student authentication, administrator management, course availability, and registration history to create a complete university course management system.

APPENDIX A – HTML CODE SNIPPETS

The following snippets represent important HTML components used in the project.

A.1 Navigation

```
<nav>
   <a href="#home">Home</a>
   <a href="#courses">Courses</a>
   <a href="#registration">Registration</a>
   <a href="#contact">Contact</a>
</nav>
```

A.2 Course Table

```
<table>
   <thead>
      <tr>
         <th>Course Code</th>
         <th>Course Name</th>
         <th>Credits</th>
         <th>Course Type</th>
      </tr>
  </thead>
  <tbody>
      <tr>
         <td>ITA0401</td>
         <td>Web Technology</td>
         <td>4</td>
```

```html
            <td>Core</td>
        </tr>
    </tbody>
</table>
```

A.3 Registration Form

```html
<form id="registrationForm">
    <label for="regNo">Register Number</label>
    <input type="text" id="regNo" required>
    <label for="studentName">Student Name</label>
    <input type="text" id="studentName" required>
    <label for="email">Email</label>
    <input type="email" id="email" required>
    <button type="submit">Register</button>
</form>
```

These HTML elements provide the basic structure and content of the registration portal.

APPENDIX B – CSS CODE SNIPPETS

The following snippets demonstrate the styling and responsive design used in the project.

B.1 Course Grid

```css
.course-grid {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 20px;
```

```
}
```

B.2 Navigation Using Flexbox

```
nav {
    display: flex;
    justify-content: center;
    align-items: center;
    gap: 25px;
}
```

B.3 Hover Effect

```
.course-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 8px 20px rgba(0, 0, 0, 0.15);
}
```

B.4 Responsive Design

```
@media (max-width: 768px) {
    .course-grid {
        grid-template-columns: 1fr;
    }
    nav {
        flex-direction: column;
        gap: 10px;
    }
}
```

These CSS techniques help create an attractive interface and ensure that the portal works properly on different screen sizes.

APPENDIX C – JAVASCRIPT CODE SNIPPETS

The following snippets demonstrate the main interactive features of the application.

C.1 Course Object

```
const courses = [
  {
    code: "ITA0401",
    name: "Web Technology",
    credits: 4,
    type: "Core"
  },
  {
    code: "ITA0402",
    name: "Database Management",
    credits: 4,
    type: "Core"
  }
];
```

C.2 Credit Calculation Function

```
function calculateCredits(selectedCourses) {
  let totalCredits = 0;
  selectedCourses.forEach(course => {
    totalCredits += course.credits;
  });
  return totalCredits;
```

```
}
```

C.3 Form Validation

```
registrationForm.addEventListener("submit", function(event) {
    event.preventDefault();
    const email = document.getElementById("email").value;
    if (!email.includes("@")) {
        alert("Please enter a valid email address.");
        return;
    }
    console.log("Registration attempt:", email);

});
```

C.4 Dynamic Registration Summary

```
summary.innerHTML = `
    <h3>Registration Summary</h3>
    <p>Student Name: ${studentName}</p>
    <p>Register Number: ${regNo}</p>
    <p>Department: ${department}</p>
    <p>Total Courses: ${selectedCourses.length}</p>
    <p>Total Credits: ${calculateCredits(selectedCourses)}</p>
`;
```

C.5 Debugging

During development, console.log() was used to inspect selected courses and identify errors.

```
console.log("Selected Courses:", selectedCourses);
```

```
console.log("Total Credits:", calculateCredits(selectedCourses));
```

This helped identify a property mismatch between credit and credits. The property was corrected to credits, allowing the credit calculation to work correctly.